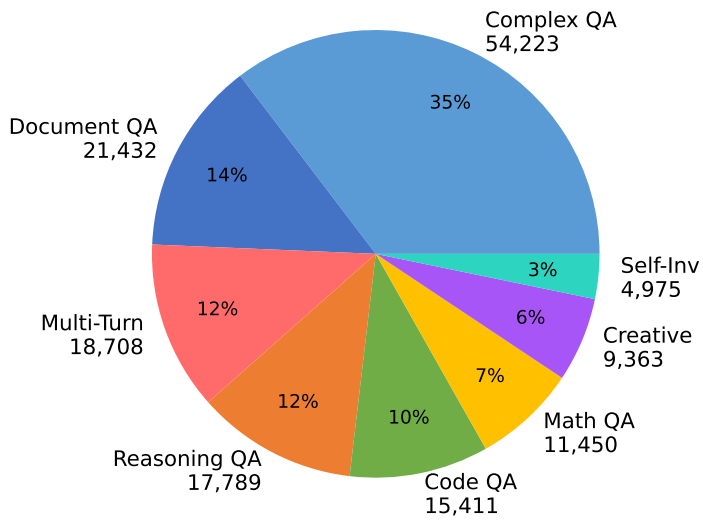
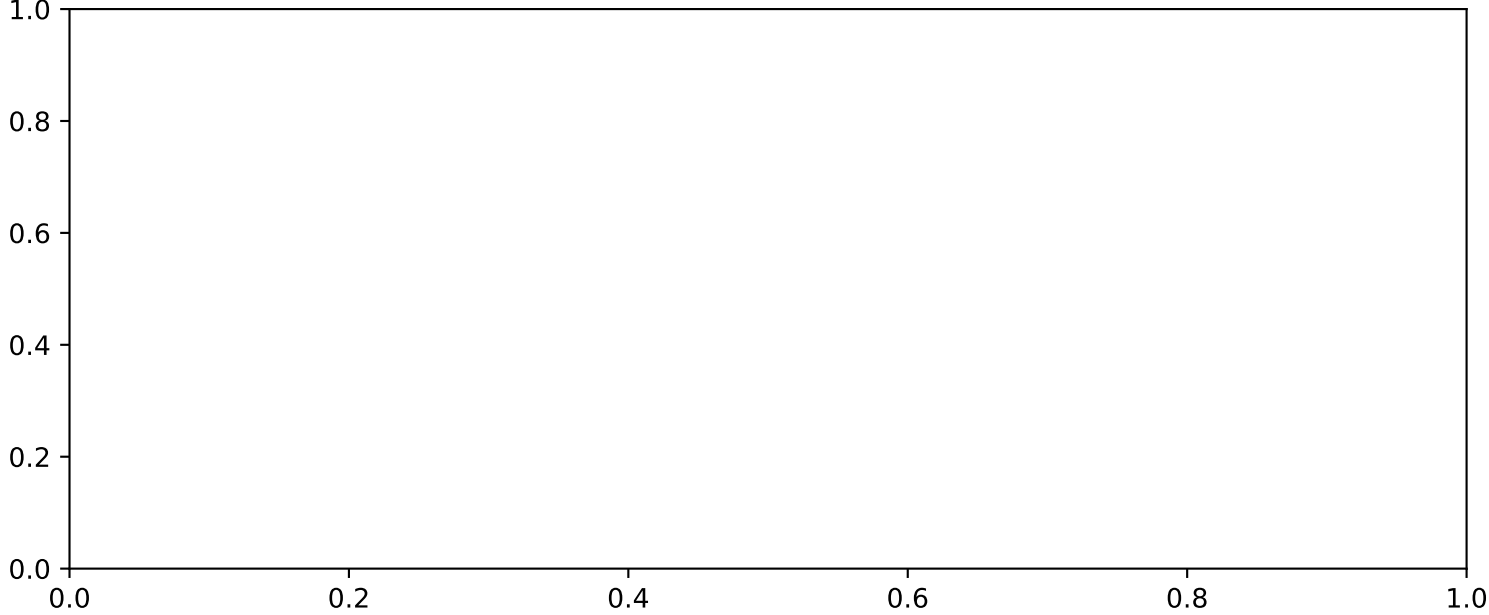


FANNO-Dev: Diversity-Verified Large-Scale Instruction Data Synthesis

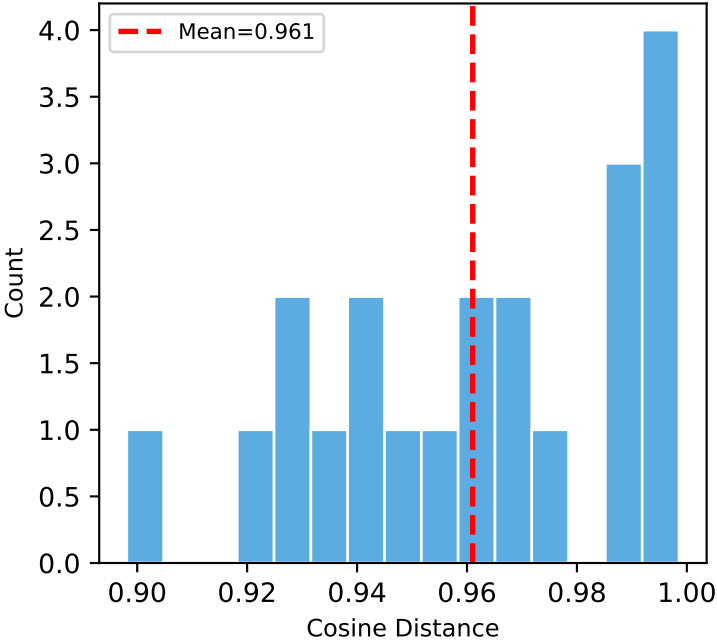
153,351 Cleaned Samples
from 8 GPT-4o Pipelines



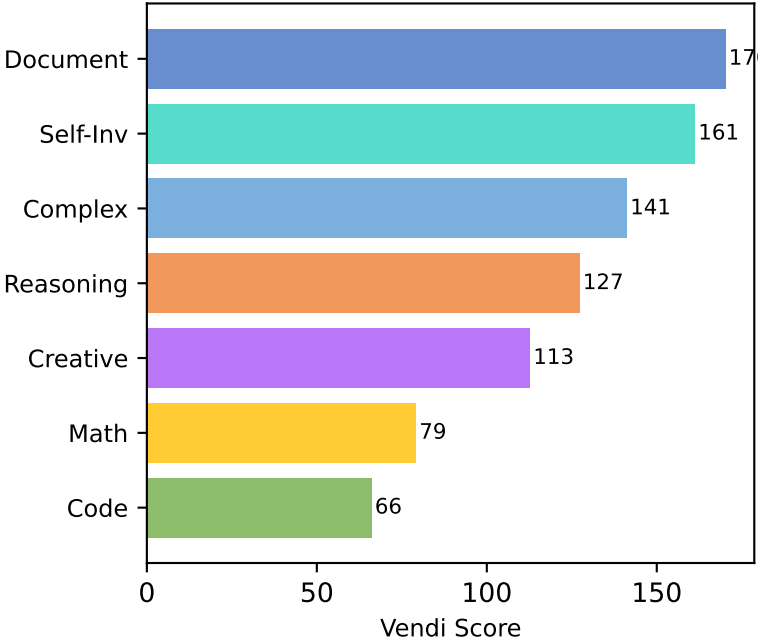
Diversity Scaling Law (R^2=0.996)



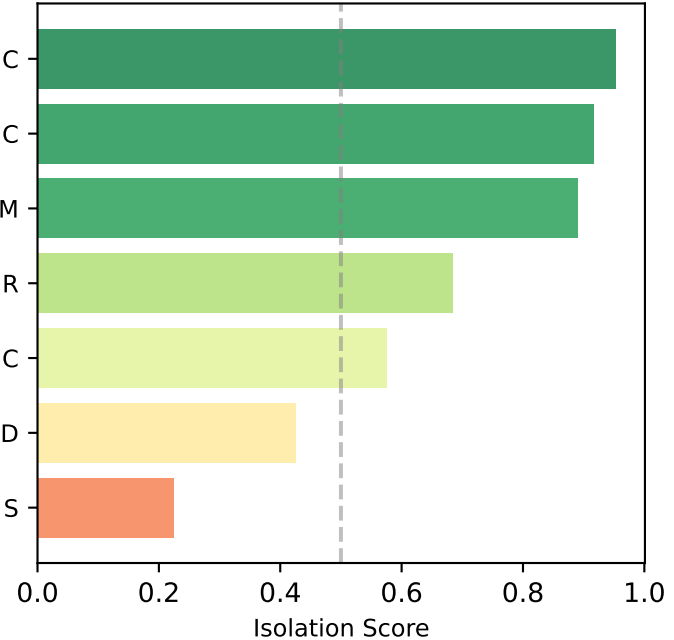
Cross-Source Distance
(avg=0.961)



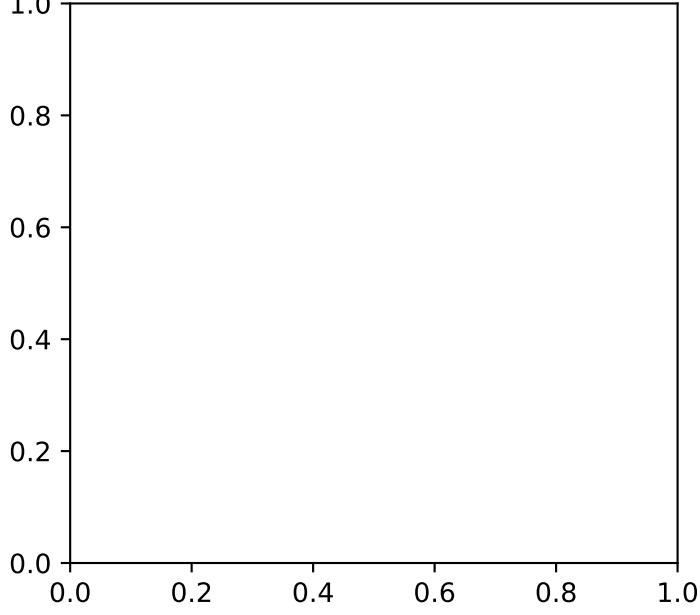
Per-Source Diversity



Source Isolation
(KNN K=20)



Selection Strategies



FANNO-Dev Key Results

- > 153,351 cleaned samples (211K raw, 27.4% rejected)
- > Vendi Score: 182.75 (15K embedding sample)
- > Scaling: $V(N)=55.4(1-\exp(-N/745))+109$ ($R^2=0.996$)
- > Cross-source distance: 0.961 (near-orthogonal pipelines)
- > 2,297 unique domains (25 question types)
- > Cost: ~\$2,245 total (\$15.58 per 1K clean samples)
- > K-Center-Greedy: +33% diversity (at N=500 vs random)
- > Self-Inversion: universal amplifier (in all 20 clusters, 0.224 isolation)
- > 0 benchmark contamination (33K+ instances checked)
- > 8 synthesis pipelines via GPT-4o (17 Azure endpoints)

Diversity Density
(oversample high-density sources)

